

Replication Plan

Physics and Chemistry from Parsimonious Representations: Image Analysis via
Invariant Variational Autoencoders

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Auto-generated Replication Blueprint

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Contents

1	Paper Overview	2
2	Detailed Workflow	2
2.1	Phase 1: Data Preparation	2
2.2	Phase 2: iVAE Architecture	2
2.3	Phase 3: Training	2
2.4	Phase 4: Latent Space Analysis	3
2.5	Phase 5: Validation and Figure Reproduction	3
3	Datasets	3
4	Models, Tools, and Software	3
4.1	Hardware Requirements	4

Paper Overview

This paper presents invariant variational autoencoders (iVAEs) for learning disentangled, physically meaningful latent representations from microscopy and spectroscopy images. The approach builds translational and rotational invariance into the VAE architecture, enabling the model to separate physical degrees of freedom (e.g., composition, strain, defect concentration) from nuisance variables (position, orientation). The authors apply the method to scanning probe microscopy (SPM) and electron microscopy data from materials science, demonstrating that the latent space organizes according to physical parameters.

Detailed Workflow

Phase 1: Data Preparation

1.1 Obtain microscopy image datasets.

- Simulated STEM or SPM images with known physical parameters (for validation).
- Experimental microscopy images from the authors or public repositories.

1.2 Preprocess images.

- Normalize intensities, crop/resize to uniform dimensions.
- Extract patches (sub-images) centered on atomic positions or grid-sampled.
- Create train/validation/test splits.

Phase 2: iVAE Architecture

2.1 Implement the encoder.

- Convolutional layers extracting features from image patches.
- Rotational invariance: use circular harmonics, steerable CNNs, or data augmentation.
- Translational invariance: global average pooling or spatial transformer.
- Output: mean μ and log-variance $\log \sigma^2$ of the latent distribution.

2.2 Implement the decoder.

- Transposed convolutions reconstructing the image from the latent code.

2.3 Loss function.

- Reconstruction loss (MSE or BCE) + KL divergence regularization.
- Optional: β -VAE weighting for disentanglement.

Phase 3: Training

3.1 Train the iVAE on the image patch dataset.

3.2 Hyperparameter tuning: latent dimension, β , learning rate, architecture depth.

3.3 Monitor reconstruction quality and KL divergence.

Phase 4: Latent Space Analysis

- 4.1 **Visualize the latent space** (2D projections via t-SNE or UMAP).
- 4.2 **Correlate latent dimensions** with known physical parameters (composition, strain).
- 4.3 **Latent space traversals**: vary one latent dimension and decode to visualize the learned transformation.
- 4.4 **Clustering in latent space** to identify distinct phases or structures.

Phase 5: Validation and Figure Reproduction

- 5.1 Reproduce latent space organization plots.
- 5.2 Reproduce reconstruction examples.
- 5.3 Reproduce quantitative metrics (disentanglement scores, reconstruction error).

Datasets

Dataset / Source	Type	Location / Notes
Simulated microscopy images	Synthetic	Generated via multislice or analytical models
Experimental SPM/STEM data	From authors	Check paper supplementary or ORNL data repositories
Code repository data	Bundled	https://github.com/ (check paper for specific repo)
dSprites (baseline benchmark)	Public	https://github.com/deepmind/dsprites-dataset

Models, Tools, and Software

Tool / Library	Version	Purpose / URL
PyTorch	≥ 1.10	Deep learning framework
NumPy / SciPy	≥ 1.21	Numerical operations
Matplotlib	≥ 3.4	Visualization
scikit-learn	≥ 1.0	Clustering, dimensionality reduction
UMAP-learn	latest	UMAP visualization https://umap-learn.readthedocs.io/
AtomAI	latest	Deep learning for microscopy (ORNL) https://github.com/pycroscopy/atomai

Tool / Library	Version	Purpose / URL
e2cnn	latest	Equivariant CNNs for rotational invariance https://github.com/QUVA-Lab/e2cnn
Jupyter	latest	Notebook environment

Hardware Requirements

- GPU with ≥ 8 GB VRAM for training.
- ≥ 16 GB RAM.